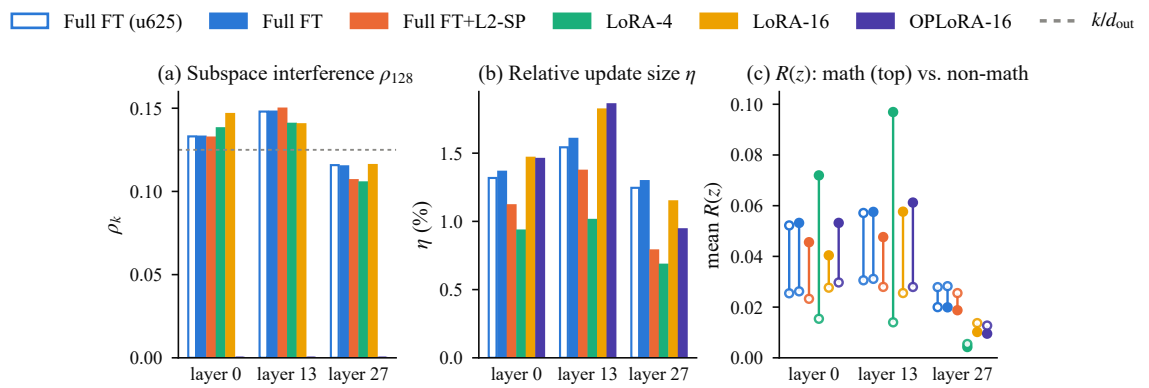
Full FT (u625)
Full FT
Full FT+L2-SP
LoRA-4
LoRA-16
OPLoRA-16
$k/d_{\text{out}}$
(a) Subspace interference $\rho_{128}$
$\rho_k$
0.15
0.10
0.05
0.00
layer 0
layer 13
layer 27
(b) Relative update size $\eta$
$\eta$ (%)
1.5
1.0
0.5
0.0
layer 0
layer 13
layer 27
(c) $R(z)$: math (top) vs. non-math
mean $R(z)$
0.10
0.08
0.06
0.04
0.02
0.00
layer 0
layer 13
layer 27